

DaTra: Data Transformation Semantics as a Symmetric Monoidal Category on Atlas Presheaves

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Abstract

This work formalizes the semantics of the Datra programming-language project in Lean. Data domains are organized into atlases: structured, eventually stable families of pages encoding how data is partitioned and related. Data transformations are modeled as presheaves on the resulting atlas categories, yielding a topos-theoretic semantics. Stable transformations are then assembled over atlas federations. A federation retains both the tagged inputs of an atlas merge and the atlas produced by that merge; forgetting the tags recovers an ordinary atlas. Horizontal sum extends by Day convolution to a symmetric monoidal category.

1 Dominions

Definition 1.1 (The Category of Dominions). The **Category of Dominions**, denoted $\mathbf{Dom}$, has as its objects sets whose cardinality is at most ω_0 , equivalently sets admitting an injection into ω_0 , and as its morphisms set-theoretic functions.

Definition 1.2 (The Domanial Embedding Functor). The **Domanial Embedding Functor**, denoted $\mathbf{Emb} : \mathbf{Dom} \rightarrow \mathbf{Set}$, sends each dominion to its underlying set.

2 Domanial Insertions

Definition 2.1 (The Category of Domanial Insertions). The **Category of Domanial Insertions**, denoted $\mathbf{DomIns}$, has dominions as objects and monomorphisms in $\mathbf{Dom}$ as morphisms.

Definition 2.2 (The Category of Codomanial Insertions). The **Category of Codomanial Insertions**, denoted $\mathbf{CoDomIns}$, is the opposite category $\mathbf{DomIns}^{\text{op}}$.

3 Chains

Definition 3.1 (Chain). A **Chain** is a totally ordered thin category, indexed skeletally by an ordinal strictly smaller than $\omega_0^{\omega_0}$.

Definition 3.2 (The Spine). The **spine** is the chain with ω_0 objects.

4 Consolidations

Definition 4.1 (The Category of Consolidations). The **Category of Consolidations**, denoted $\mathbf{Con}$, has chains as objects and point-surjective functors between their thin categories as morphisms.

Definition 4.2 (The Category of Coconsolidations). The **Category of Coconsolidations**, denoted $\mathbf{CoCon}$, is the opposite category $\mathbf{Con}^{\text{op}}$.

Definition 4.3 (Consolidation Transport). The **Consolidation Transport Functor** is denoted $\mathbf{ConTra}$. It is a functor from $\mathbf{Con}$ to $\mathbf{Set}$ that sends a chain to its object set and a consolidation to its underlying set-theoretic surjection.

5 Folios and Paginations

Definition 5.1 (Folio). A **folio** is a functor $F : S \rightarrow \mathbf{CoCon}$, where S is the spine, such that $F(0)$ is the singleton chain.

Definition 5.2 (Page Elements). For a folio F , its **category of page elements**, denoted $\mathbf{El}(F)$, has as objects pairs (m, k) , where m is a page and k is an object of the chain $F(m)$. An arrow $(m, k) \rightarrow (n, l)$ is an arrow $f : m \rightarrow n$ in the opposite spine for which the consolidation induced by F transports k exactly to l . Equivalently,

$$\mathbf{El}(F) = \int (\mathbf{ConTra} \circ F^{\text{op}}).$$

Definition 5.3 (The Category of Paginations). The **Category of Paginations**, denoted $\mathbf{Pag}$, has as objects a folio Fo together with its associated category of page elements $El = \mathbf{El}(Fo)$. For $W : \mathbf{Pag}$, write W_{Fo} for its folio and W_{El} for its page-element category. A morphism $T : X \rightarrow Y$ in $\mathbf{Pag}$ is a functor $T : X_{El} \rightarrow Y_{El}$.

For illustration, let $X_H := \mathbf{ConTra} \circ X_{Fo}^{\text{op}}$ and define Y_H likewise. If an arrow f of X_{El} has source $x = (m, i)$, then T acts schematically as follows:

$$\begin{array}{ccc} x = (m, i) & \xrightarrow{f} & (n, X_H(f)(i)) \\ T \downarrow & & \downarrow T \\ (m', i') & \xrightarrow{T(f)} & (n', Y_H(T(f))(i')). \end{array}$$

6 Atlases

Definition 6.1 (The Category of Atlases). The **Category of Atlases**, denoted $\mathbf{Atl}$, has as objects pairs (Pa, Da) , where $Pa : \mathbf{Pag}$ and $Da : Pa_{El} \rightarrow \mathbf{DomIns}$ is a functor, equivalently a presheaf $Da : Pa_{El}^{\text{op}} \rightarrow \mathbf{CoDomIns}$. For the folio Pa_{Fo} , a page m , and distinct objects k, k' of the chain $Pa_{Fo}(m)$, the pullback in Da of the arrows induced by $(m \rightarrow 0)(k)$ and $(m \rightarrow 0)(k')$ is empty.

For $X : \mathbf{Atl}$, write X_{Pa} and X_{Da} for its two components, and abbreviate $X_{Fo} := (X_{Pa})_{Fo}$ and $X_{El} := (X_{Pa})_{El}$.

For every $n \in \mathbb{N}$, the **n th atlas page** of X is the chain $X_{Fo}(n)$ on its full spine. An object k of this chain is a **page cell**, written (n, k) , and the dominion carried by that cell is $X_{Da}(n, k)$.

There is an integer w such that, for every $w' > w$, the consolidation from page w' to page w induced by Pa_{Fo} is the identity and, for every object k of $Pa_{Fo}(w)$, the corresponding arrow $Da((w' \rightarrow w)(k))$ is the identity. The least such integer is the **cardinality of the atlas**, denoted $|A|$.

A morphism $T : X \rightarrow Y$ is a pair (T_{Pa}, T_{Da}) , where $T_{Pa} : X_{El} \rightarrow Y_{El}$ is a functor and $T_{Da} : X_{Da} \Rightarrow Y_{Da} \circ T_{Pa}$ is a natural transformation. Thus, for every $f : x \rightarrow y$ in X_{El} , the following square commutes:

$$\begin{array}{ccc} X_{Da}(x) & \xrightarrow{X_{Da}(f)} & X_{Da}(y) \\ (T_{Da})_x \downarrow & & \downarrow (T_{Da})_y \\ Y_{Da}(T_{Pa}(x)) & \xrightarrow{Y_{Da}(T_{Pa}(f))} & Y_{Da}(T_{Pa}(y)). \end{array}$$

Finally, if $x = |X|$, $x' > x$, and r is an object of the final chain $X_{Fo}(|X| - 1)$, then $T_{Pa}(x', r) = T_{Pa}(x, r)$ and $(T_{Da})_{(x', r)} = (T_{Da})_{(x, r)}$.

Definition 6.2 (Extent of an Atlas). The **extent** of A is $\text{Ex}(A) = A_{Da}(0, 0)$. Since objects of DomIns are dominions, $\text{Ex}(A) : \text{Dom}$ for every $A : \text{Atl}$.

Definition 6.3 (Territory of an Atlas). Let M be the objects of the final chain $A_{Fo}(|A| - 1)$. The **territory** is the indexed family $\text{Ter}(A) : M \rightarrow \text{Dom}$ given by $\text{Ter}(A)(k) = A_{Da}(|A| - 1, k)$.

Definition 6.4 (n th Region of an Atlas). The n th **region** of A is $\text{Ter}(A)(n)$, with indexing beginning at 0.

7 Transposals and Traversals

Definition 7.1 (The Category of Atlas Transposals). The **Category of Atlas Transposals**, denoted AtlTrap , is the wide subcategory of Atl whose morphisms $F : X \rightarrow Y$ have point-injective object maps F_{Pa} .

Definition 7.2 (The Category of Atlas Traversals). The **Category of Atlas Traversals**, denoted AtlTrav , is the wide subcategory of AtlTrap whose morphisms satisfy the following conditions. For $F : X \rightarrow Y$, let $x = (m, i)$ and $y = (m', j)$, put $p = \min(m, m')$, write $F_{Pa}(x) = (n, i')$ and $F_{Pa}(y) = (n', j')$, and put $p' = \min(n, n')$. Let $c_{m,p}^X$ and $c_{m',p}^X$ denote the consolidations induced by the folio X_{Fo} , and define the analogous maps for Y . If

$$c_{m,p}^X(i) < c_{m',p}^X(j),$$

then

$$c_{n,p'}^Y(i') < c_{n',p'}^Y(j').$$

Furthermore, if $q \in |X|$ and $t \in X_{Da}(q)$ is in the image of a final region—that is, there exist $k \in |\text{Ter}(X)|$ and $l \in \text{Ter}(X)(k)$ whose image under $X_{Da}((|X| - 1) \rightarrow q)(k)$ is t —then this property is preserved by F .

Definition 7.3 (The Category of Stable Atlas Traversals). The **Category of Stable Atlas Traversals**, denoted StaAtlTrav , is the wide subcategory of AtlTrav whose morphisms $F : X \rightarrow Y$ satisfy $F_{Pa}(0, 0) = (0, 0)$.

8 Maps and Cartography

Definition 8.1 (The Category of Atlas Maps). The **Category of Atlas Maps**, denoted AtlMap , is the full subcategory of atlases for which every element of the extent is in the image of a final region.

Definition 8.2 (The Atlas Map Inclusion Functor). The **Atlas Map Inclusion Functor** $\text{AtlMapInc} : \text{AtlMap} \rightarrow \text{Atl}$ is the canonical inclusion.

Definition 8.3 (The Category of Atlas Traversal Maps). The **Category of Atlas Traversal Maps**, denoted AtlTravMap , is the full subcategory of AtlTrav whose objects are Atlas Maps.

Definition 8.4 (The Atlas Traversal Map Inclusion Functor). The canonical inclusion is denoted $\text{AtlTravMapInc} : \text{AtlTravMap} \rightarrow \text{AtlTrav}$.

Lemma 8.5 (Cartography Lemma). *The inclusion AtlTravMapInc has a right adjoint, denoted the **Charting Functor** $\text{Chr} : \text{AtlTrav} \rightarrow \text{AtlTravMap}$.*

Proof sketch. On an object $X : \text{AtlTrav}$, the functor Chr sends X to the universal arrow from AtlTravMapInc to X . Write $X' = \text{AtlTravMapInc}(\text{Chr}(X))$, together with $H : X' \rightarrow X$, with X' terminal among objects having this property. The solution is the DaTra map satisfying $\text{Ter}(X') \cong \text{Ter}(X)$. It is unique because AtlTrav preserves orders and terminal because H_{Pa} is faithful.

9 Coalitions

Definition 9.1 (The Coalizing Functor). The **Coalizing Functor**, denoted

$$\text{Coa} : \text{StaAtlTrav} \longrightarrow \text{DomIns},$$

is the extent of the charted atlas: $\text{Coa} = \text{Ex} \circ \text{Chr}$.

Definition 9.2 (The Coalition of an Atlas). The **coalition** of an atlas X is the extent of its chart, regarding X as an object of the wide category of stable traversals.

Definition 9.3 (The Domanial Inclusion Functor). The **Domanial Inclusion Functor** sends a dominion to the atlas of cardinality 1 whose coalition is that dominion. Write $\text{DomInc} : \text{DomIns} \rightarrow \text{StaAtlTrav}$ for this functor. It is left adjoint to Coa : for $X : \text{DomIns}$ and $Y : \text{StaAtlTrav}$, naturally in X and Y ,

$$\text{Hom}_{\text{StaAtlTrav}}(\text{DomInc}(X), Y) \cong \text{Hom}_{\text{DomIns}}(X, \text{Coa}(Y)),$$

and $\text{Coa}(\text{DomInc}(X)) \cong X$.

10 Atlas Federations

Definition 10.1 (Atlas Merge). The **Atlas Merge** is the object operation

$$\text{AtlMerge} : \text{Ob}(\text{Atl}) \times \text{Ob}(\text{Atl}) \longrightarrow \text{Ob}(\text{Atl}).$$

For atlases X and Y , the extent of $\text{AtlMerge}(X, Y)$ is the disjoint union of their extents. Both atlases are evaluated on their full spines: page $n + 1$ of the merge is the tagged, componentwise combination of page n of X and page n of Y .

Definition 10.2 (Atlas Federation). An **Atlas Federation** is the data carried by an atlas merge: a countable tagged family of atlas objects together with a merge presentation whose evaluation is the resulting atlas. The category of Atlas Federations is denoted AtlFed . A morphism consists of a map of tags and, at each source tag, a stable atlas traversal to its selected target tag. There is a canonical object-level forgetful operation $U : \text{Ob}(\text{AtlFed}) \rightarrow \text{Ob}(\text{Atl})$ that discards the component tags and retains the resulting atlas.

Definition 10.3 (The Empty Atlas). The **Empty Atlas** is $\text{AtlI} = \text{DomInc}(\emptyset)$, also viewed as the empty Atlas Federation, whose tag family is empty and whose stored result is AtlI .

Definition 10.4 (The Atlas Horizontal Sum Bifunctor). The **Atlas Horizontal Sum Bifunctor**, denoted $\text{AtlHorSum} : \text{AtlFed} \times \text{AtlFed} \rightarrow \text{AtlFed}$, has object action

$$U(\text{AtlHorSum}(F, F')) = \text{AtlMerge}(U(F), U(F')).$$

On tags, $\text{AtlHorSum}(F, F')$ is their disjoint tagged union; its stored merge presentation records the displayed atlas merge. Thus the horizontal sum preserves both the provenance of the two inputs and the atlas produced by merging them. The empty federation represents AtlI . Given stable traversals between the components of two pairs of federations, horizontal sum preserves the left and right tags and applies the traversals componentwise. Stability fixes the common origin, so the component arrows remain stable atlas traversals.

Lemma 10.5 (Horizontal Lemma). *The Atlas Horizontal Sum Bifunctor AtlHorSum exists.*

Proof. Identity arrows act identically on every tag, composition holds componentwise, and every component arrow is required to lie in StaAtlTrav . The federation structure supplies the associator, unitors, and braider by reassociation, deletion of the empty tag family, and tag swapping, respectively.

Definition 10.6 (The Atlas Braider). The **Atlas Braider** $\text{AtlBrd}_{F,F'} : \text{AtlHorSum}(F, F') \rightarrow \text{AtlHorSum}(F', F)$ swaps the left and right component tags. Hence $\text{AtlBrd}_{F,F'} \circ \text{AtlBrd}_{F',F} = \text{Id}$.

Definition 10.7 (The Atlas Associator). The **Atlas Associator**, denoted

$$\text{AtlAsoc}_{F,F',F''} : \text{AtlHorSum}(\text{AtlHorSum}(F, F'), F'') \longrightarrow \text{AtlHorSum}(F, \text{AtlHorSum}(F', F'')),$$

flattens the nested federation and retags its three sides using the canonical reassociation $((x + y) + z) \leftrightarrow (x + (y + z))$. Its data component is the matching reassociation of the three extent summands.

Definition 10.8 (The Atlas Unitors). The **Atlas Left Unitor** $\text{Lu} : \text{AtlHorSum}(\text{Atll}, F) \rightarrow F$ and the **Atlas Right Unitor** $\text{Ru} : \text{AtlHorSum}(F, \text{Atll}) \rightarrow F$ remove the empty tag family.

11 Data Transformations

Definition 11.1 (The Category of Data Transformations). The **Category of Data Transformations**, also called the category of **Data Transformation Sets** or **DaTra Sets**, is the presheaf category

$$\text{DaTra} = [\text{Atl}^{\text{op}}, \text{Set}].$$

As a presheaf category, it is a topos.

Definition 11.2 (Navigation). A **navigation** of $D : \text{DaTra}$ is a monomorphism $\text{Nav} : \text{Yo}(A) \rightarrow D$ for some atlas A .

Definition 11.3 (Expedition). An **expedition** is a navigation represented by an Atlas Map.

Definition 11.4 (Data Transformation Maps). The **Category of Data Transformation Maps**, denoted DaTraMap , is the full subcategory of DaTra Sets all of whose navigations are expeditions.

12 Stable Data Transversals

Definition 12.1 (Stable Data Transversals). The **Category of Stable Data Transversals**, denoted StaDaTrav , is the presheaf category on Atlas Federations whose component arrows lie in StaAtlTrav . Correspondingly, each stable data transversal has an underlying DaTra presheaf.

Definition 12.2 (The Empty Map). The **Empty Map**, denoted I , is the Day unit represented by the empty Atlas Federation:

$$I = \text{Yo}(\text{Atll}).$$

Definition 12.3 (The Horizontal Sum Bifunctor). The **Horizontal Sum Bifunctor**, denoted $\text{HorSum} : \text{StaDaTrav} \times \text{StaDaTrav} \rightarrow \text{StaDaTrav}$, or in infix notation by $\boxplus : \text{StaDaTrav} \times \text{StaDaTrav} \rightarrow \text{StaDaTrav}$, is the Day convolution extension of stable atlas horizontal sum. Because the indexing morphisms are arrows of StaAtlTrav , its result and its arrow action land in StaDaTrav by construction.

Definition 12.4 (Stable Data Transversals Monoidal Category). The **Stable Data Transversals Monoidal Category**, denoted StaDaTravMon , is

$$\text{StaDaTravMon} = (\text{StaDaTrav}, \boxplus, I).$$

Its tensor is Day convolution on the Atlas Federation. Its braider, associator, and unitors are based on the Atlas Braider, Atlas Associator, and Atlas Unitors of Definitions 10.6–10.8, respectively.